

Lab sheet 04**Submission Due: 19.06.2024****Title:** Introduction to JavaScript.**Aims:**

- Running JavaScript in different ways
- Getting knowledge about data types in JavaScript
- Getting understand about objects and functions in JavaScript
- Getting familiar with event handling in JavaScript

Tasks:

- JavaScript placement
- Variable & Data Types
- Introduction to Objects and Functions in JavaScript.
- Exploring Event Handling in JavaScript

Activities :

1. JavaScript Placement in HTML file
 - I. Script in <head> section

```
<head>
  <title>JavaScript Placement in HTML file</title>

  <script>
    function sayHello() {
      alert("Hello World")
    }
  </script>
</head>
```

- II. Script in <body> section

```
<body>
  <input type = "button" onclick = "sayHello()" value = "Say
Hello" />

  <script>
    function sayHello() {
      alert("Hello World")
    }
  </script>
</body>
```

III. Script in an external file

```
// script.js
function sayHello() {
    console.log("Hello World")
}
```

```
<!-- index.php -->
<!DOCTYPE html>
<html lang="en">
<head>
    <title>JavaScript Placement in HTML file</title>
    <script src = "script.js" ></script>
</head>
<body>
    <input type = "button" onclick = "sayHello()" value = "Say
Hello" />
</body>
</html>
```

2. Variables & Data Types

I. Variables

```
<script>
    // Without using any keywords.
    // Using the 'var' keyword.
    // Using the 'let' keyword.
    // Using the 'const' keyword.

    var name;
    marks;
    _age;
    var gpa = 3.4;

    let reg_no = "ICT000";
    const DEP = "Faculty of Technology";

</script>
```

II. Data Types

```
<script>
    // String
    var str1 = "Hello";

    // Integer
    var num1 = 45;

    // Floating point number
    var float1 = 78.45;
```

```
// Boolean
var bool1 = true;

// Undefined
let udef1 = undefined;

// Null
let var1 = null;

// Object
const obj = {
  animal: "Hourse",
  legs: 4,
  hourseColor: "Brown"
}

// Array
const colors = ["Brown", "red", "pink", "Yellow", "Blue"];

// Date
let date = new Date();

</script>
```

3. Object and functions

I. Function

```
// script.js

// function name(parameter1, parameter2, parameter3) {
//   code to be executed
// }

function addition(number1, number2) {
  return number1 + number2;
}

<p id="cal_function"></p>

<script>
  let sum = addition(5,6);
  document.getElementById("cal_function").innerHTML = sum;
</script>
```

II. Object

```
// Creating Object
const house = { address: "Main street, B City", bedrooms: 4,
bathroom: 2 };

// Create empty object and add properties
const house2 = {};

house2.address = "Main street, B City";
house2.bedrooms = 4;
house2.bathroom = 2;

// Create object with new keyword
const house3 = new Object();
house3.address = "Main street, B City";
house3.bedrooms = 4;
house3.bathroom = 2;
```

III. Object Method

```
const house = {
  address: "Main street, B City",
  bedrooms: 4,
  bathroom: 2,
  getAddress: function () {
    return this.address;
  }
};
```

```
<h5>Address : </h5>
<p id="address"></p>
<script>
  let address = house.getAddress();
  document.getElementById("address").innerHTML = address;
</script>
```

IV. Object Constructor

```
function Person(first_name, last_name, age) {
  this.first_name = first_name;
  this.last_name = last_name;
  this.age = age;
  this.getFullName = function() {
    return `${this.first_name} ${this.last_name}`;
  };
}
```

```
<h5>Person : </h5>
<p id="person"></p>
<script>
    let person1 = new Person("Oliver","George", 30);
    document.getElementById("person").innerHTML =
person1.getFullName();
</script>
```

4. Event handling

```
function handleSelectChange() {
    const selectElement = document.getElementById("mySelect");
    const selectedValue = selectElement.value;

    document.getElementById("selectedValue").textContent = `You
selected: ${selectedValue}`;
}
```

```
<select id="mySelect" onchange="handleSelectChange()">
    <option value="Apple">Apple</option>
    <option value="Orange">Orange</option>
    <option value="Mango">Mango</option>
</select>
<p id="selectedValue"></p>
```

Event Name	Description
onchange	Fires when the value of an element changes
onclick	Triggers when the user click on an element
onmouseover	Occurs when the mouse moves over an element
onmouseout	Fires when the mouse moves away from an element
onkeydown	Triggers when the user presses a key down
onload	Fires when the entire web page finishes loading

Discussion:

- Difference between let and const
- JavaScript keywords